

## Selection Sort — Definition

**Selection Sort** is a simple sorting algorithm that repeatedly finds the **smallest element** from the unsorted part of an array and places it at the beginning of that part.

## Algorithm

1. Start with the first element of the array.
2. Assume it is the smallest element.
3. Compare it with all the remaining elements.
4. Find the actual smallest element.
5. Swap the smallest element with the first unsorted element.
6. Move to the next position.
7. Repeat the process until the entire array is sorted.

Output:

```
Enter number of elements: 5
Enter the elements:
2
4
6
8
9
Sorted array: [2, 4, 6, 8, 9]

...Program finished with exit code 0
Press ENTER to exit console.[]
```

## Time and Space Complexity of Selection Sort

- **Best Case:**  $O(n^2)$
- **Average Case:**  $O(n^2)$
- **Worst Case:**  $O(n^2)$
- **Space Complexity:**  $O(1)$

**Reason:** Selection sort uses two nested loops to find the minimum element, so it takes  $O(n^2)$  comparisons. It sorts **in-place**, requiring only a small amount of extra memory, so its space complexity is  $O(1)$ .

